Who is on the other side of your trade?

What everyday traders pay on a public-wallet exchange, and which rules would protect them

Rent or toxicity: strategic exploitability of learning market makers with public wallet information

Deepu Pradeep Miguel Betran Menz

October 27, 2026

6.S974 Games, Learning, and Security, Workshop 1
Example 4: an algorithmic market with adaptive agents

Good morning. We are Deepu and Miguel, Example 4: an algorithmic market with adaptive agents. The title is the question an everyday trader never gets to ask: who is on the other side of my trade, and what are they taking from me? On the exchange we study, every trade carries a public wallet address, so for once that question can be answered, and we will show what the professionals do with the answer. The technical version of the question is in brackets: when the professionals quote a wide price, is it a rent they are collecting, or a defence against toxic flow? By the end you will know why those two are impossible to tell apart from the outside, and what that means for the people trading against professionals. Ten minutes: two on why, three on the model, four on what we found, one on what is next.

2026: the year retail leverage went wrong at national scale

−44%
KOSPI, June 19 to July 29

1.2 million
retail accounts margin-called: 3.4% of all adults

$39 billion
retail losses on single-stock leveraged ETFs

“Retail” means everyday people

- 92% of the holders were individuals, not firms
- 62% of the accounts: people aged 20 to 39
- trading on a phone app, with borrowed money, against professionals they cannot see

President Lee Jae-myung, on X, October 14, 2026

살아야지요. 어떻게든 사셔야 합니다. 힘드시면 109번 전화를 하십시오.

사실 길, 살아야 하는 이유를 함께 찾아드릴 것입니다.

“You must live. Somehow, you must live. If it is hard, call 109. We will find, together, a way and a reason to live.”

Sources: Bloomberg, Reuters, Goldman Sachs, Citi (via Seoulz, “KOSPI Crash 2026”); Maeil Shinmun, Oct 22, 2026; x.com/Jaemyung_Lee/status/2110145832511225984. 109 is Korea’s suicide-prevention line.

Why we care. This summer the Korean stock market fell forty-four percent in six weeks, and July 28 was the worst single day on record. What made it a national event rather than a market event is who was holding the risk. The word for that is retail, and retail means everyday people: not banks, not funds. Ninety-two percent of the holders of these products were individuals. By mid-July 1.2 million retail accounts had been margin-called, which is three point four percent of every adult in the country, and three hundred thousand were force-liquidated. Citi puts retail losses on single-stock leveraged ETFs alone at thirty-nine billion dollars, and sixty-two percent of those accounts belonged to people in their twenties and thirties. Students, office workers, people with ordinary jobs, trading on a phone app with borrowed money against professionals they could not see. On October 14 the President of Korea posted this on X: you must live; if it is hard, call 109, the suicide-prevention line. That is the level at which this stopped being a finance story. Our project is about the gap between what one of those people can see and what the professional opposite them can see, and we think closing that gap is a financial-literacy problem as much as a market-design one.

What “leverage” is, and why a 5% move can take everything

You put in $1,000. At 20× leverage you control a $20,000 position.



yours: $1,000 **borrowed from the exchange**: $19,000

price +5%

position worth $21,000

you doubled your money: +$1,000

price −5%

position worth $19,000: exactly the loan

you lost everything: −$1,000

the exchange closes your position automatically (**liquidation**); in a crash, millions of these happen at once and push the price further down

A 2× ETF (Korea) does this for you at 2×. On a crypto perpetuals exchange the multiplier is a **slider**, up to 50×.

The yield is visible on the way in. The risk, and who is on the other side, is not.

Since nobody here needs to know this for a living: leverage is trading with borrowed money. You put in a thousand dollars; at twenty times leverage the exchange lends you nineteen thousand more and you control a twenty thousand dollar position. If the price goes up five percent, the position is worth twenty-one thousand; you pay back the nineteen, you keep two, you doubled your money. That is the yield people see. If the price goes down five percent, the position is worth nineteen thousand, which is exactly the loan. Your thousand is gone. The exchange does not wait for you: it closes the position automatically, which is called liquidation, and in a crash millions of these happen at once and push the price down further, which liquidates the next person. In Korea the product was a two-times ETF, a fund that does this for you at two times. On a crypto perpetuals exchange the multiplier is a slider that goes to fifty. The yield is visible on the way in; the risk, and who is on the other side taking your money, is not. That second part is what we study.

These products are moving toward US retail; Korea should be the lesson

In the US, right now

- **CLARITY Act** (crypto market structure): House 294 to 134, July 2025; one vote short in the Senate, Sept 15, 2026; revivable
- **CFTC**, Aug 2026: rules to let exchanges offer leveraged crypto trading to Americans; filed Sept 17
- **Hyperliquid** blocks US users today; its CEO is discussing a CFTC path

In Korea, after the crash

- new listings halted; minimum deposit tripled; exposure capped at 20% of assets. All of it *after* the losses.

On a public-wallet venue, every trade you make is on the tape

fill buy 0.5 BTC @ 67,412.0
taker `0x7f3a...c91e` maker `0x21b0...4d7a`
public, forever; the professionals read it

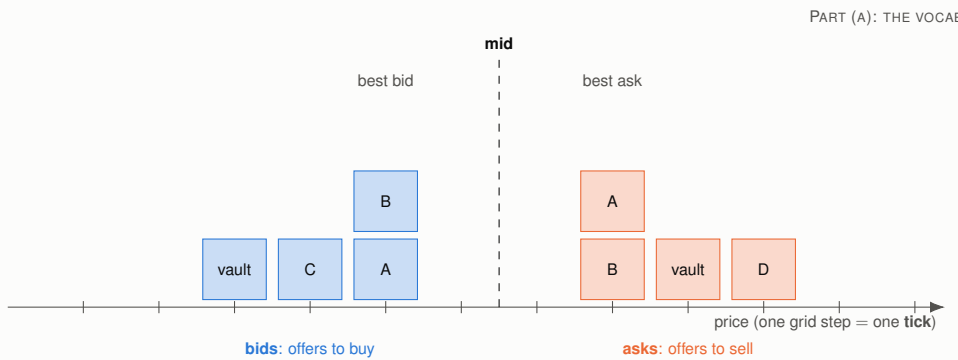
What an everyday trader would need to know

- what a trade really costs
- who is on the other side
- whether their identity is used against them
- which venue rules protect them in stress

Korea's protections came after the losses. We measure these four things now, on the real venue and in a model where the rules can be changed, so the protections can come first.

The products that did this in Korea are moving toward US retail, and the legislative machinery is in motion. The CLARITY Act, the crypto market-structure bill, passed the House two hundred ninety-four to one hundred thirty-four in July 2025; it failed a Senate procedural vote forty-nine to fifty on September 15 this year, but the no vote that sank it was cast specifically to preserve a path to bring it back. Separately, in August the CFTC chairman announced rules that would let exchanges offer leveraged crypto trading to Americans under CFTC oversight, and that rulemaking was filed in September. Hyperliquid, the venue we study, blocks US users today, but its CEO has been publicly discussing exactly that path. And on this kind of venue there is one more thing: every trade you make carries your wallet address on a public tape, forever, and the professionals on the other side read it. We will show that they do. Korea's response was real: new listings halted, the minimum deposit tripled, exposure capped at twenty percent of assets. But every one of those came after the losses. We think what happened in Korea should be read as a lesson, and the protections should come first. For that, someone has to measure what an everyday trader would need to know: what a trade really costs beyond the quoted price; who is on the other side and earns from them; whether their public identity is used against them; and which venue rules protect the uninformed in a stress episode, and which quietly make them pay. That is what this project measures, on the real venue and in a model where the rules can be changed.

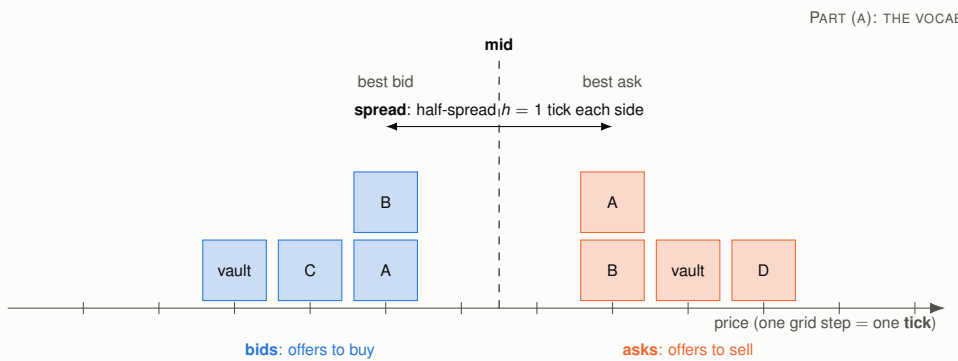
An exchange is a list of standing offers: the order book



Makers post the offers (the boxes). **Takers** accept them.
Each box carries a public **wallet** address.

Vocabulary, because the audience has not seen this before. An exchange is a list of standing offers, the order book. Price lives on a grid; one grid step is a tick. On the left, in blue, are bids: offers to buy at that price. On the right, in orange, are asks: offers to sell. The people who post these standing offers are market makers; each box is one maker's order and on Hyperliquid each box carries the maker's public wallet address. The dashed line is the mid, halfway between the best bid and the best ask. [click] The distance between best bid and best ask is the spread; we write the half-spread h in ticks, here one tick each side. [click] A taker is someone who accepts an offer instead of posting one. T buys one unit: he hits the best ask, and that is a fill for maker B, because B was first in line at that price and A was behind. A bigger order takes A's too and walks to the next price, and that walking is most of what a trade really costs. What becomes public: the fill itself, buy, price, both wallets, immediately. What does not: whether B made or lost money on it. That only shows up as the price moves afterwards, which is the next slide.

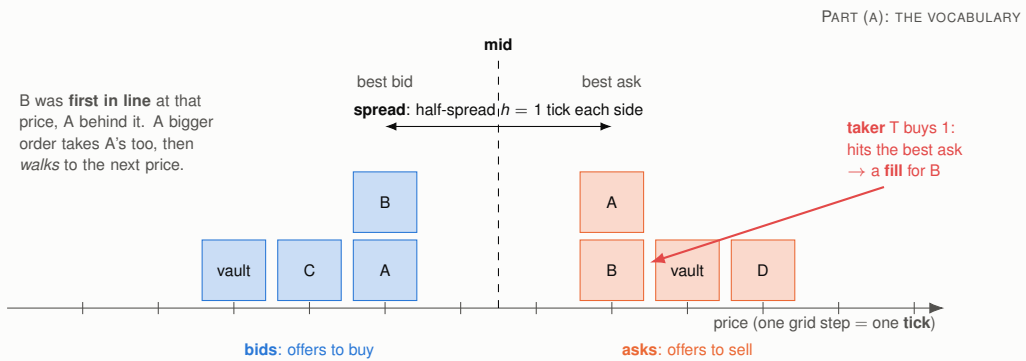
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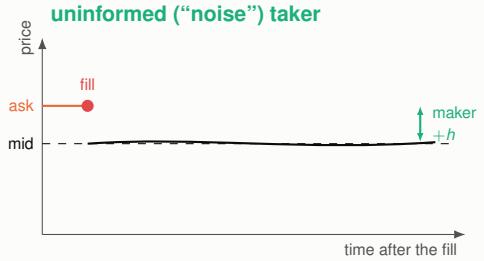


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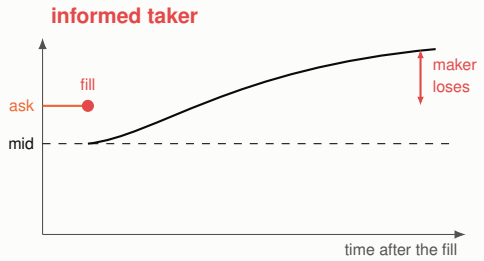
Public, at once: buy, price, wallet T, wallet B.
Public, only later: whether B gained or lost on it.

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Did the maker win? Look at the price afterwards: the markout



price does not move; the maker keeps the half-spread



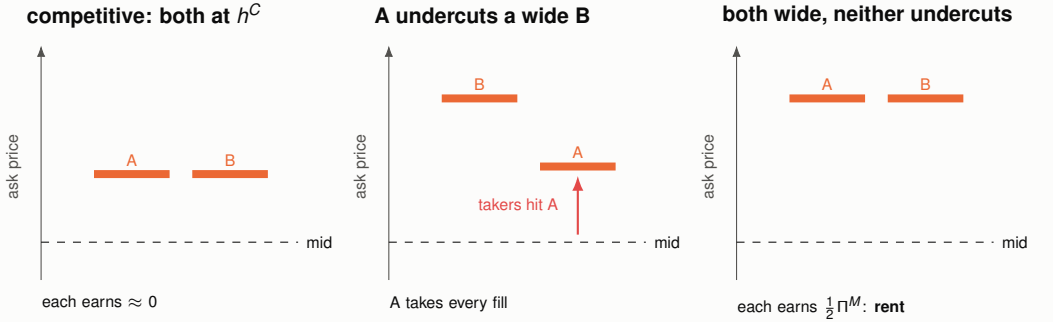
the taker knew; the price runs past the fill price

markout = mid a few seconds later − fill price, signed for the maker

toxic flow: a stream of fills with negative markouts (*adverse selection*)

How do we know whether the maker won on a fill? We look at where the price goes afterwards. On the left, the taker was uninformed, a noise trader; the price sits where it was, and the maker sold above the mid, so it keeps the half-spread. On the right the taker knew something; the price runs up past the price the maker sold at, and the maker has sold too cheap. That signed difference, mid a few seconds later minus the fill price from the maker's point of view, is the markout. It is the single measurement everything in our data work rests on. A stream of fills with negative markouts is what practitioners call toxic flow, and economists call adverse selection. A maker's whole business is being in line for the left-hand fills and not the right-hand ones.

Rent: what makers earn by not undercutting each other

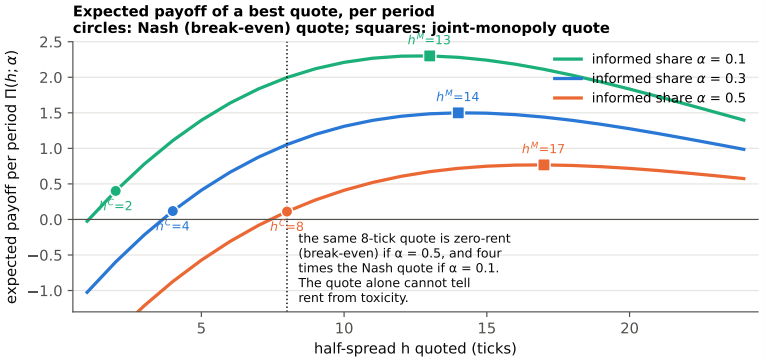


h^C: the **competitive** quote, break-even. Wider pays, but invites undercutting.

Learning makers end up in the right panel *without talking*: **tacit collusion**. The extra is the **rent**.

The other word in the title: rent. Three pictures of the same two makers' ask prices. Left: both quote the competitive half-spread h-C, where what you earn from noise takers just covers what you lose to informed ones; each earns about zero. Middle: B quotes wide and A undercuts by a tick; takers always hit the tighter price, so A takes every fill and B gets nothing. That is why a wide quote is fragile. Right: both quote wide and neither undercuts. Each takes half the fills at the wide price and earns half the monopoly profit. The difference between the right panel and the left is the rent. The striking fact, from Colliard, Foucault and Lovo and replicated by us, is that independent learning algorithms end up in the right panel without ever communicating. Each learns that undercutting today gets punished tomorrow, and stops. That is tacit collusion, and the rent is what the uninformed taker pays for it.

The puzzle: on the tape, rent and toxicity look identical

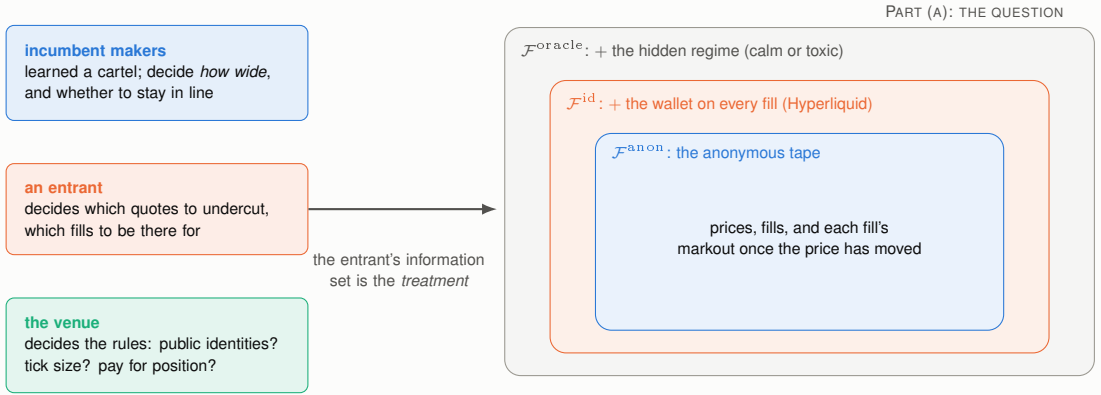


A wide quote is a **rent** . . . or the correct charge for **toxic flow**.

Which curve you are on is decided by α, the share of informed takers. Only outcomes, or *who* trades, reveal it.

Here is the puzzle. This is the maker's expected profit per period as a function of its half-spread, one curve per level of toxicity alpha, the fraction of takers who are informed. Each curve has a break-even point on the left, h-C, and a peak, h-M. More toxic flow pushes the whole curve down and both points to the right. So if you watch a market and see wide quotes, you cannot tell whether the makers are colluding on a calm market, the blue curve, or correctly charging for a toxic one, the red curve. Rent and toxicity produce the same tape. The only ways to tell are to look at outcomes, the markouts, or to look at who is trading. And that second route exists only because wallets are public.

Three players, and what each one gets to know



Who plays. Three players. The incumbents, in blue, are the makers who have already learned a cartel; they decide how wide to quote and, on a one-tick book, whether to stay in the queue. The entrant, in orange, is the adversary: a new maker who decides which quotes to undercut and which fills to be there for. The venue, in green, sets the rules: are identities public, how coarse is the tick, can a maker pay for queue position. The right side is the information structure, nested. The anonymous tape, which every exchange in the world shows: prices, fills, and eventually each fill's markout. Add the wallet on every fill and you have what Hyperliquid shows. Add the hidden regime, whether the market is currently calm or toxic, and you have an oracle that we use only as an upper bound. The entrant's information set is the treatment: we run the same attacker with each of the three and the difference is what identity is worth.

What we want to learn

RQ1 Value of identity

What is a public wallet worth to an entrant, beyond the tape, and through which mechanism?

RQ2 Exploitability

How much of a learned cartel's rent can an adversary take or destroy?

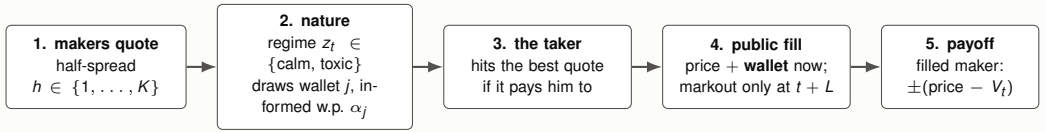
RQ3 One-tick books

When the grid pins the spread, where does rent live, and which rules decide who holds the toxic flow?

Today: the rent is fully exploitable; identity's value is a lag story; and on the real book the makers who earn are the ones avoiding the toxic wallets.

Three questions. One: what is a public wallet worth to an entrant beyond what the anonymous tape already tells it, and through what mechanism. Two: how much of a learned cartel's rent can an adversary take for itself or destroy, and does a bigger rent mean a bigger target. Three: on Hyperliquid's major books the spread is pinned at one tick by the grid, so the quote cannot carry rent; where does the rent live then, and which venue rules decide who ends up holding the toxic flow. Why this is a game and not an optimisation: the entrant's weapon, undercutting or cherry-picking fills, is exactly what the incumbents' learned reaction function is tuned to punish, and the venue decides what each side is allowed to know. The three headlines today: the rent is fully exploitable; identity's value is a story about lag; and on the real book the makers who earn the spread are the ones who are missing from the toxic wallets' fills.

One period of the game



Incumbents: tabular Q-learners, state = (own last h, rival's last h); trained, then **frozen**.

Entrant: information set $\mathcal{F}^{\text{anon}} \subset \mathcal{F}^{\text{id}} \subset \mathcal{F}^{\text{oracle}}$; a Bayesian filter estimates the next taker's α.

Payoff curve, closed form:
$\Pi(h;\alpha) = (1-\alpha)\, 2x\bar{\Phi}(x/\sigma_L) - \alpha\mu e^{-x/\mu}$,
$x = h\theta$.

The model, one period. Step one: every maker commits a half-spread h on the tick grid. Step two: nature draws a regime, calm or toxic, from a persistent Markov chain, then a wallet from a population of fifty; each wallet has its own probability alpha-j of being informed, and in the toxic state the active wallets are the toxic ones. Step three: the taker hits the best quote if it pays him to, which for an informed taker means exactly when the maker will lose. Step four: the fill is public at once, price and wallet, but its markout, the value V-t the maker traded against, is only revealed L arrivals later. That lag L is the key parameter. Step five: the filled maker's payoff is price minus value. One hundred periods make an episode; the regime chain and the wallets persist across episodes. The incumbents are tabular Q-learners whose state is their own last quote and the best rival's; they are identity-blind by construction; we train them twenty thousand episodes and then freeze them, so a cartel is a fixed reaction function. The entrant's information set is the treatment, and a Bayesian filter turns whatever it sees into an estimate of the next taker's alpha. And because the market is Glosten-Milgrom, the expected payoff of a quote against informed share alpha is closed form, which is what makes rent measurable rather than estimated.

Benchmarks are closed-form, so “rent” is measured, not estimated

$h^C = \min\{h : \Pi(h) \geq 0\}$	**Nash** (competitive)
$h^M = \arg\max_h \Pi(h)$	**joint monopoly**
$C = \dfrac{\Pi_{\text{cartel}} - \Pi^C}{\Pi^M - \Pi^C} \in [0,1]$	**rent index**
$L_{\mathrm{BR}} = \Pi_{\text{cartel}} - \Pi_{\text{under attack}}$	**loss under attack**
$\kappa_A = L_{\mathrm{BR}}/(\Pi_{\text{cartel}} - \Pi^C)$	**rent destroyed**

At $\bar{\alpha} = 0.3$: $h^C = 4$, $h^M = 14$, rent $\Pi^M - \Pi^C = 1.38$.

Why a cartel is attackable: it is a *reaction function* that does not chase every undercut. That restraint is the rent, and the attack surface.

One-tick (queue) mode: spread pinned; actions REST / OUT / REST-AND-PAY, first in first out. A fill at h^C: $+1.10$ calm, -2.26 toxic.

Because the payoff curve is closed form, the benchmarks are exact. h-C is the smallest break-even quote, the Nash outcome of one-shot competition: any wider quote is profitably undercut. h-M is the joint-monopoly quote. The rent index C puts a frozen cartel between zero, Nash, and one, monopoly; the cartels we train sit at 0.7 to 0.9. Under an adversary A, L-B-R is what the cartel loses, and kappa-A is that loss as a share of the rent it had; kappa above one means the attacker destroyed more than the rent. All our results are in rent units, divided by Pi-M minus Pi-C, so cartels at different toxicity levels are comparable. A worked example at alpha 0.3: two incumbents at fourteen ticks split 1.50 per period. An entrant at thirteen takes every fill and earns 1.49, they earn zero. If they chase, the entrant goes to twelve, and the chase ends at h-C equals four, where everyone earns 0.12. A learned cartel is a reaction function that does not chase every time; that restraint is the rent, and it is also the attack surface, because the attacker can learn exactly when it chases and when it does not. The queue mode models the one-tick book: the spread is pinned, so the only decisions are rest, step out, or pay for priority, and the fill queue is first in first out. At break-even a fill is worth plus 1.10 in a calm spell and minus 2.26 in a toxic one, so all the value is in which fills you take.

The prototype, in one command

`python experiments/demo.py`

1. benchmarks h^C, h^M, $\Pi(h)$
2. the shipped cartel alone: $C = 0.86$
3. three adversaries enter
4. the best response, period by period

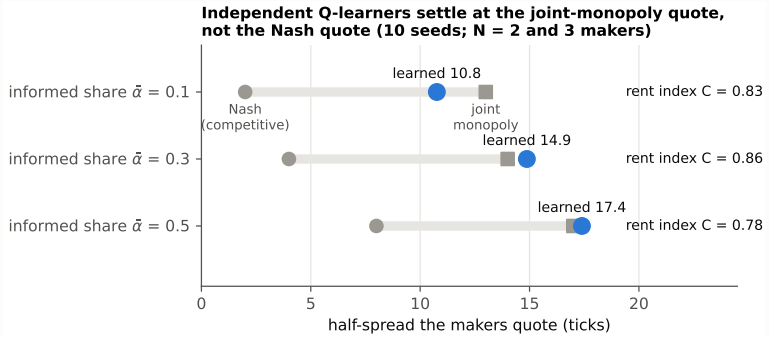
`github.com/pradeepu191/wargame`
59 tests; every result logged with its command.

```
Entry (200 episodes each, same random numbers):
  adversary                          cartel loses  entrant earns  kappa_A
  undercut (one tick inside)               0.41           0.33     0.48
  markout (undercut if rent inferred)      0.47           0.41     0.55
  exact best response                      0.94           1.09     1.10

Trace: exact best response vs the cartel (quotes in ticks)
   t  inc0  inc1  entrant   taker          filled   payoff
   0    15    24       12   noise    buy  entrant   +1.21
   1    23    16       13   noise    buy  entrant   +7.86
   2    20    14       13   no trade
   5    21    19       14   informed buy  entrant   -5.77
   7    18    23       14   noise    buy  entrant  +11.79
```


[0:15] Everything you are about to see runs from one command in seconds: benchmarks, the cartel alone, three adversaries entering under the same random numbers, and the best response traced period by period. Fifty-nine tests, every number in the report logged with the command that produced it.

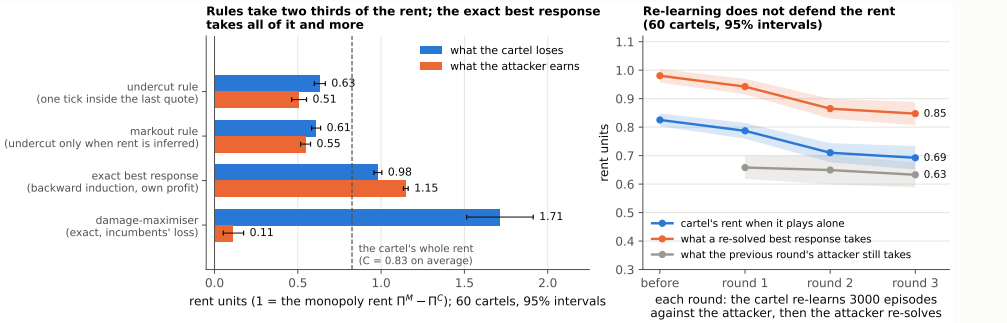
Evidence 1: learners coordinate above the Nash point



$C = 0.71$ to 0.87 in every cell. In 55 to 83% of states undercutting would pay *this period*; the learner forgoes it 96 to 100% of the time.

[0:30] First, the thing we attack exists. Independent Q-learners settle at a rent index of 0.7 to 0.9 in every cell of a grid over the number of makers, learning rate and initialisation; that replicates Colliard, Foucault and Lovo. And we can see the mechanism: in most states undercutting would pay this period, and the learner almost never does it. It has learned that the rival punishes. That restraint is the rent, and it is also the attack surface.

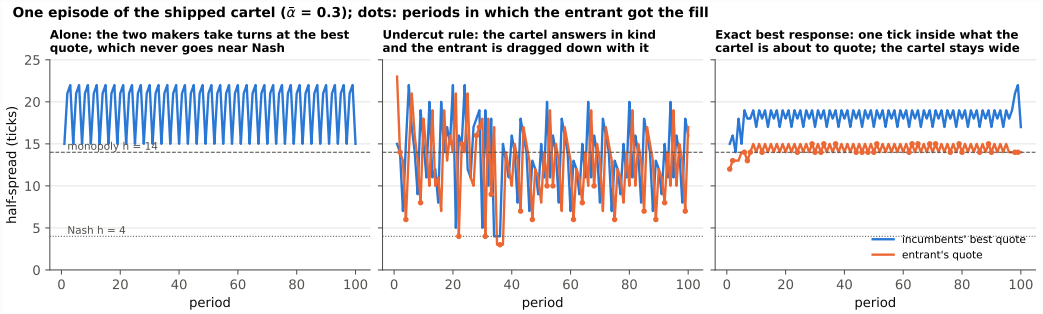
Evidence 2: the rent is fully exploitable



Hand-written rules take two thirds. The **exact best response** takes it all, in 60 of 60 cartels: $\kappa_A = 1.1$ to 1.3.
The damage-maximiser destroys 2 to 3$\times$ the rent at no cost.

[0:50] How much can an adversary take? Two hand-written rules, undercut by a tick, or undercut when the markouts say there is rent, take about two thirds, flat across a hundred and eighty cartels. The exact best response is different in kind. A frozen cartel is a deterministic reaction function, so the entrant's problem is a dynamic program we solve by backward induction. It takes more than the whole rent in sixty of sixty cartels and earns the monopoly profit for itself. And flipping the objective to damage, it steers the cartel inside break-even and destroys two to three times the rent while taking no fills at all. In a double oracle where the cartel re-learns against it, the re-solved attacker takes it again; the numbers are in the backup.

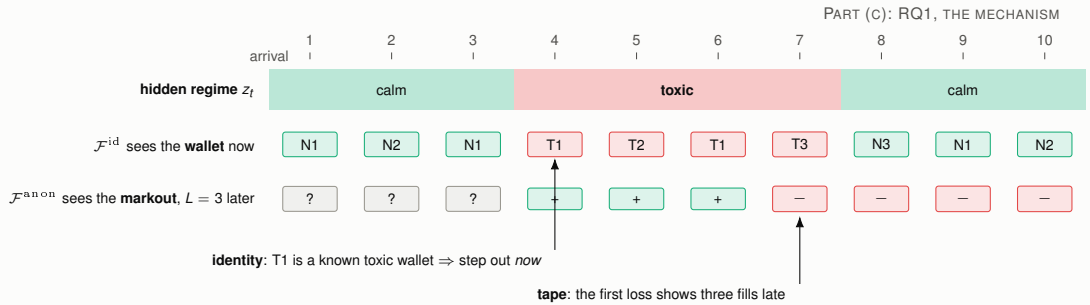
Steering, period by period



Left: the cartel alone rotates at wide quotes. **Middle**: a naive undercut starts a price war. **Right**: the best response holds the cartel wide and takes the fills.

[0:30] What that looks like. Left: the cartel alone rotates at wide quotes and never goes near Nash. Middle: a naive undercut starts a price war and both sides slide to break-even. Right: the best response never provokes the war. It quotes one tick inside the reaction it has predicted; the cartel's learned answer is to stay wide; and the entrant takes nearly every fill at the monopoly price. The cartel's own restraint does the work.

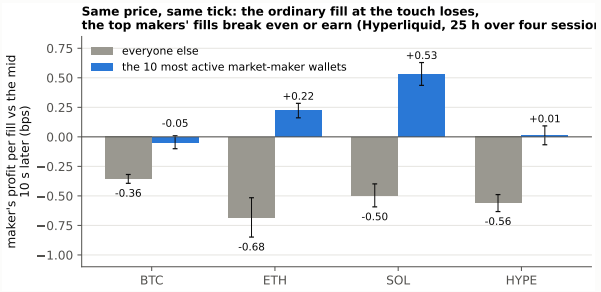
Why identity has value: a spell of toxic flow, seen two ways



Identity is worth what you lose in the L fills before the tape catches up: measured, zero at $\bar{\alpha} \leq 0.3$, growing with L at $\bar{\alpha} = 0.5$ (backup).

[0:40] Now RQ1, what a wallet is worth beyond the tape. Informed trading comes in spells, and the toxic wallets are the ones active in a spell. The regime turns toxic at arrival four. With identities, the fourth fill carries a wallet you have seen lose you money before, so you step out now. The anonymous tape learns a fill's markout L arrivals later; with L equal to three, the first minus sign appears at arrival seven. So identity is worth what you lose on those L fills: measured, it is zero where even toxic flow leaves an undercut profitable, alpha up to 0.3, and grows with the lag at alpha 0.5, up to the lifetime of a spell. On Hyperliquid that lag is real: the outcome of a fill is knowable only once the price has moved; the wallet is on the fill at once.

Evidence 3, Hyperliquid: the price cannot move, and someone still earns



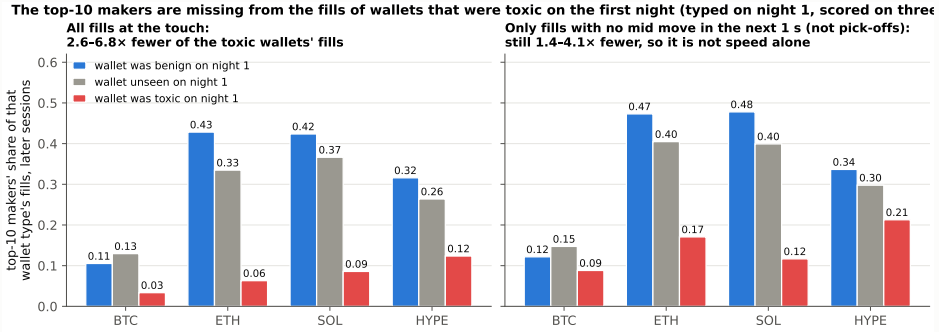
Four sessions, 25 h, 490k scored fills.

- **The spread is one tick**, essentially always
- **The ordinary fill loses**: -0.2 to -0.4 bps in 10 s
- **The top makers earn** $+0.3$ to $+1.0$ bps more, *at the same price*; rank persists (corr. 0.5 to 0.8)

Rent lives in *which fills you get*, not in the quote.

[0:45] The real venue. Four sessions, twenty-five hours, four hundred and ninety thousand scored fills. Three facts. The spread is one tick essentially always, so rent cannot live in the price. The ordinary fill loses: a maker filled at the touch collects the tick and loses 0.2 to 0.4 basis points to the mid within ten seconds. And the ten most active makers, chosen on activity and scored on outcome, earn 0.3 to 1.0 basis points more per fill at the same price, session after session. So on the real book, rent lives in which fills you get.

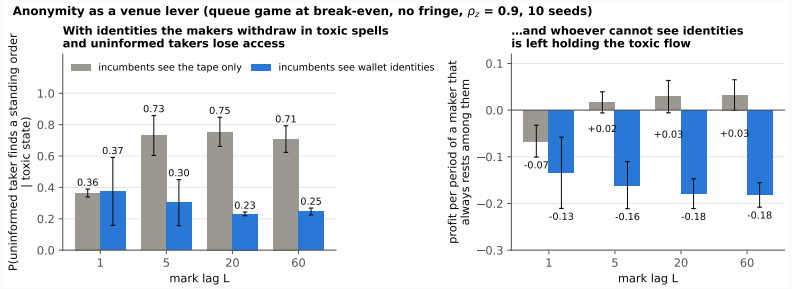
Evidence 4: how they do it is identity, not the tape



They never withdraw after a stressful spell (30 s to 5 min). They are absent from the toxic *wallets'* fills: typed on night 1, scored on three later sessions.

[0:45] How do they get the good fills? Not by reading the tape: they never withdraw after a stressful spell at any timescale from thirty seconds to five minutes. By identity. We typed every taker wallet on the first night by the markout it inflicted, then on three later sessions measured each maker's share of fills against toxic versus benign wallets. The top makers hold eleven to forty-three percent of the benign wallets' fills and three to twelve percent of the toxic ones; the gap is the same in calm and stressed tape and survives on fills where the mid never moved. That is RQ1's mechanism on the real book: the makers who earn are the ones missing from the toxic wallets' fills, and someone else is holding those.

Evidence 5: the levers decide who holds the toxic flow, not whether rent survives



Anonymity: a transfer from makers to uninformed takers in stress; identity the reverse.

Priority fee: the venue collects 2 to 4× the rent; the identity-blind fringe pays.

Makers' own levers: only an identity-aware floor helps, and it leaves the adversary 75 to 80%.

[0:30] Who holds them is the venue's choice, in our one-tick queue model. Anonymity moves money from makers to uninformed takers in stress; identity moves it back. A priority fee lets the venue collect two to four times the makers' rent, paid by the fringe that cannot see identities. The makers' own defences mostly cost profit and buy nothing. So the rules decide who holds the toxic flow, not whether the rent survives, and the party holding it is the one that cannot see.

Plan for Workshop 2

Attacks

A **public-information attacker**: identify the reaction function from the tape, then solve. Then both sides learning (PSRO).

Designs

Venue and maker levers on one frontier: profit vs the worst 5% under attack, plus takers' access in stress.

Data

A month of recording: the identity-selection gap by coin, hour and regime; which *action* carries the selection.

Your view: is a frozen reaction function the right "realistic attacker", or should the cartel adapt continuously?

[0:25] Three things for Workshop 2. Attacks: an attacker with public information only, which identifies the cartel's reaction function from the tape and then solves it, and then both sides learning. Designs: every venue and maker lever on one frontier, profit against the worst five percent under attack, with takers' access in stress so that robust cannot mean withdrew. Data: the recorder has run continuously since October 25 and will run a month, so we can cut the identity gap by coin, hour and regime. The question we would most like your view on: is a frozen reaction function the right realistic attacker, or should the cartel adapt continuously?

If you remember three things

1 Rent is not defensible

Solve the cartel's reaction function and you take the monopoly profit. Re-learning does not get it back.

2 Identity is a lag story

A wallet reveals a toxic spell before the price does. Worth zero when the tape is instant, more with the lag.

3 The real book agrees

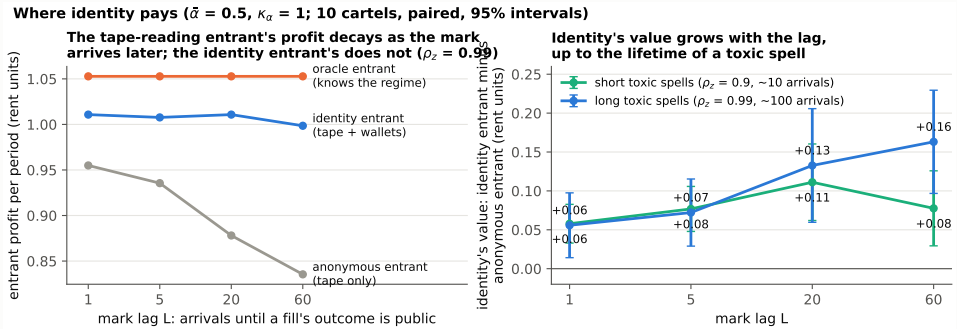
On a one-tick book rent is which fills you get. The makers who earn avoid last night's toxic wallets.

`github.com/pradeepu191/wargame`

Recorder running continuously since October 25.

[0:25] Three things. The rent is not defensible: solve the cartel's reaction function and you take the monopoly profit, and re-learning does not get it back. Identity is a lag story: a wallet reveals a toxic spell before the price does. And the real book agrees: on a one-tick book the rent is which fills you get, and the makers who earn are the ones avoiding last night's toxic wallets. Code, results and a findings log are on GitHub. Thank you.

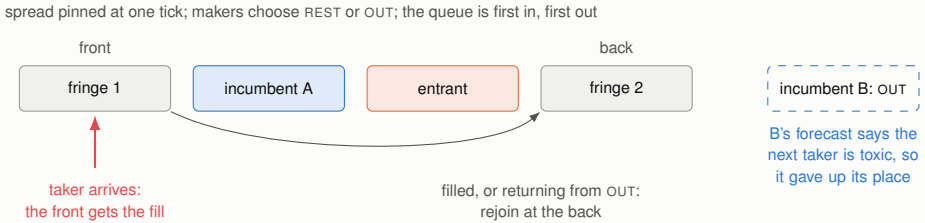
Backup: what identity is worth, measured



The proposal's mechanism (the same wallet trades again) is worth nothing: repeats are 2 to 5%. The lag mechanism survives: zero where $\bar{\alpha} \leq 0.3$, growing with L up to the spell's lifetime.

Measured. Our original proposal's mechanism, that the same wallet trades again so you can learn it, is worth nothing on the data: a wallet repeats within the window two to five percent of the time. The mechanism that survives is the lag story from the previous slide. The value of identity is zero wherever even the toxic state leaves an undercut profitable, alpha up to 0.3, because then you do not care who is coming. It becomes positive at alpha 0.5 and grows with the mark lag L, saturating at the spell's lifetime. So RQ1's answer is: identity is a lag story, and its value depends on the venue's reporting delay and the market's toxicity, not on recognising individuals.

Backup: the queue game, the one-tick book



A fill at h^C: $+1.10$ in a calm spell, -2.26 in a toxic one.

Who pays: whoever rests through the toxic spell, the *fringe* (retail orders, the vault).

The model for the one-tick book. The spread is pinned, so each maker's only decision is rest or step out, and those resting stand in a first-in-first-out line; the taker takes the front. A filled maker rejoins at the back; so does one that stepped out and comes back, which is the cost of stepping out. Incumbent B has stepped out because its forecast says the next taker is toxic; it gives up its place in line to avoid that fill. An identity-aware maker does this on the wallet label; a tape-reading one on markouts L late. A fill at break-even is worth plus 1.10 in a calm spell and minus 2.26 in a toxic one, so all the value is in which fills you take, and time priority rations them. Who pays? Whoever rests through the toxic spell: the fringe, which is retail orders and the vault. We also allow rest-and-pay, a priority fee that buys the front of the line, which turns the calm fills into an all-pay auction.

Backup: Hyperliquid numbers

	BTC	ETH	SOL	HYPE
one tick (bps)	0.12	0.39	0.86	0.11
taker's cost vs the mid: fills at the touch / walked (bps)	0.12 / 2.35	0.42 / 3.20	0.92 / 5.20	0.17 / 3.57
10-s maker markout, at the touch / walked (bps)	-0.41 / -0.80	-0.56 / -2.26	-0.27 / $+0.98$	-0.45 / -0.01
next fill's markout after a toxic vs a benign wallet (bps, 327 h)	-0.64	-1.01	-0.59	-0.59
10-s markout at the touch: top-10 makers / everyone else (bps)	-0.05 / -0.36	$+0.22$ / -0.68	$+0.53$ / -0.50	$+0.01$ / -0.56
per-maker rank correlation, first session vs later (n makers)	0.80 (113)	0.53 (80)	0.70 (53)	0.59 (70)
top-10 share of touch fills: benign / unseen / toxic wallets	.11 / .13 / .03	.43 / .34 / .06	.42 / .37 / .09	.32 / .26 / .12
. . . on fills with no mid move in 1 s: benign / toxic	.12 / .09	.47 / .17	.48 / .12	.34 / .21

Four book sessions over two weeks (25 h). Standard errors cluster by five-minute block; every top-10-vs-rest gap exceeds four SEs. Wallet types from the first night, scored on the three later sessions.

Backup: the double oracle and robustness, in numbers

- **Double oracle** (cartel re-learns 3000 episodes against the fixed best response, entrant re-solves; 3 rounds): the now-stale attacker's profit falls 1.2 $\rightarrow$ 0.5 (the undercut-rule level); the re-solved one takes $\geq$ 90% of the remaining rent in every cartel every round; the cartel's own rent erodes 0.83 $\rightarrow$ 0.79 $\rightarrow$ 0.71 $\rightarrow$ 0.69.
- **Damage-maximiser** ($\lambda = 0$): steers the cartel to 4 to 7 ticks, inside break-even ($h^C = 8$ to 13), by quoting above it; takes no fills, earns ≈ 0; $L_{\mathrm{BR}} = 1.3$ ($\bar{\alpha} = 0.3$), 2.7 to 2.8 ($\bar{\alpha} = 0.5$).
- **Train at $\bar{\alpha}$, test at $\bar{\alpha}'$**: cartels keep $C = 0.80$ to 0.86, $\kappa_A = 0.65$ to 0.75 (rules) on calmer markets; on a more toxic one the rent collapses ($C = 0.23$) and is more than destroyed.
- **Held-out adversaries** preserve every defence ordering (rank correlation 0.75 to 0.96).
- **Caveat we own:** the exact best response needs the cartel's reaction function (oracle knowledge of the *policy*). It is the true upper bound; the public-information attacker that identifies it is next.

Backup: the queue game and the priority fee

- Learned incumbents need a **fringe** of always-resting makers (else a toxic spell empties the book for good); tabular Q cannot learn to queue (time priority pays only after consecutive periods in line); **no-regret (EXP3)** incumbents over complete policies learn to select in a few hundred episodes.
- Against selecting incumbents identity is worth *more*: $+0.05$ to $+0.06$ per period at every lag, 5/5 seeds; the fringe's payoff falls to -0.02 to -0.10: identity-blind resting liquidity is the counterparty of the selection rent.
- **Priority fee**: every selector pays in calm periods (identity-aware incumbents 93 to 94% of calm, 8 to 9% of toxic periods); a non-paying entrant among payers gets 0% of fills; the fringe's loss triples at any fee; revenue peaks at $f = 0.5$.
- **Makers' levers** (20 cartels, 7 adversaries incl. a Q-learning one): requoting every 2 to 5 periods costs everything; randomising costs profit and buys nothing; the identity floor: worst-case profit 0.08 $\rightarrow$ 0.19 at $\bar{\alpha} = 0.5$.